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Table of Contents

List of Figures	v
List of Tables	vii
Nomenclature	ix
1 Theoretical Foundations	1
1.1 Principal Component Analysis	1
1.1.1 Derivation from Maximum Variance Formulation	2
1.1.2 Sparse PCA via Rotation	7
1.1.3 Hilbert PCA	11
1.2 Maximum Covariance Analysis	14
1.3 Neural Networks	18
1.3.1 Introduction to Neural Networks	18
1.3.2 Recurrent Neural Networks	18
1.3.3 Explainable AI Techniques	19
1.4 Gaussian Processes	19
1.4.1 Introduction to Gaussian Processes	20
1.4.2 Gaussian Processes for Regression	20
1.4.3 Spatial Modeling with Gaussian Processes	20
1.4.4 Model Implementation and Inference	20
1.5 Comparative Perspective	21
2 Dimensionality Reduction Techniques in Climate Science: Bridging Theory and Practice	23
2.1 Open-Source Tools for Scalable Climate Data Analysis	23

2.2	Challenges in Large-Scale and Multidimensional Climate Data Analysis	25
2.3	The XEOFS library: A Solution For Dimensionality Reduction in Climate Science	26
2.4	Conclusion	28
References		31
Appendix A Proofs		39

List of Figures

1.1	Comparison of the first three modes obtained from PCA vs. Varimax-rotated PCA for a synthetic data set. (Left) Original data set with three different, uncorrelated autoregressive processes. (Middle) PCA solution. (Right) Varimax-rotated PCA solution.	12
2.1	Open-source software (OSS) projects in Earth system sciences.	24
2.2	Computational performance of PCA using the XEOFs library (version 3.0.0). . .	29

List of Tables

1.1	Computational complexity of PCA algorithms	7
2.1	Overview of available methods in the XEOFS library	27

Nomenclature

Greek Symbols

π $\simeq 3.14\dots$

Acronyms / Abbreviations

AMO Atlantic Multidecadal Oscillation

BSISO Boreal Summer Intraseasonal Oscillation

ENSO El Niño–Southern Oscillation

MJO Madden-Julian Oscillation

NAO North Atlantic Oscillation

PC Principal Component

PDO Pacific Decadal Oscillation

VC Vector of Coefficients

Theoretical Foundations

nicrie: Motivation for Dimensionality Reduction: Explain why dimensionality reduction is necessary in climate science. e.g. ERA5 contains more than 9 PB of data. Discuss the need to simplify complex datasets by identifying the most significant patterns and features, which can help in understanding underlying processes and improving predictability.

1.1 Principal Component Analysis

Principal Component Analysis (PCA) is a statistical technique used to reduce the dimensionality of a dataset while preserving most of the variance. This is achieved by identifying the principal axes, which are the directions in which the data varies the most. These directions correspond to the eigenvectors of the covariance matrix of the data, with the associated eigenvalues representing the magnitude of variance along these directions. PCA is particularly useful for dimensionality reduction, data compression, feature extraction and visualization of high-dimensional datasets.

The origins of PCA can be traced back to Pearson [92] in 1901 and Hotelling [56] in 1936. It gained significant traction in meteorology through the work of Lorenz [79] in 1956, who introduced the technique under the term *empirical orthogonal function* (EOF) analysis¹. The method has since been extensively applied in meteorology and climate science to identify dominant modes of variability in the atmosphere and oceans, as discussed in numerous textbooks [e.g. 28, 46, 64, 95, 109, 123].

In this section, we derive PCA from the maximum variance formulation², as outlined by Bishop [6, pp. 561-563]. This approach involves projecting the data onto a lower-dimensional linear space, known as the principal subspace, to maximize the variance of the projected data. We will then explore the limitations of PCA, particularly in the context of climate science, and discuss common strategies to address these limitations.

¹Although EOF analysis remains the preferred term in meteorology and climate science, we will use the more commonly recognized term, PCA.

²Hotelling [56] originally proposed this formulation. An alternative derivation, known as the minimum error formulation by Pearson [92], seeks a linear projection that minimizes the average projection error, defined as the mean squared distance between the original data points and their projections. This alternative approach shows that PCA is an optimal solution for approximating data in the squared error sense.

nicrie: make link to Karhunen-Loève expansion

1.1.1 Derivation from Maximum Variance Formulation

Consider a data set represented as a complex-valued, two-dimensional $N \times D$ matrix $\mathbf{Z}$, where N denotes the number of observations (or samples) and D the number of variables (or features). We will refer to the n^{th} observation as the D -dimensional vector $\mathbf{z}_n$. For simplicity, we assume the data is centered, meaning the mean of each variable is zero, i.e. $\bar{\mathbf{z}} = \frac{1}{N} \sum_{n=1}^N \mathbf{z}_n = \mathbf{0}$. The goal is to project the data onto an M -dimensional space ($M < D$) of M latent variables while maximizing the variance of the projected data.

To start, consider a one-dimensional projection ($M = 1$). We define the direction of this space using a D -dimensional unit³ vector $\mathbf{v}_1$, such that $\mathbf{v}_1^H \mathbf{v}_1 = 1$, where $\mathbf{v}_1^H$ denotes the conjugate transpose of $\mathbf{v}_1$. The data points $\mathbf{z}_n$ are then projected onto this direction, $\mathbf{v}_1^H \mathbf{z}_n$, and the variance of the projected data is

$$\frac{1}{N-1} \sum_{n=1}^{N-1} (\mathbf{v}_1^H \mathbf{z}_n - \mathbf{v}_1^H \bar{\mathbf{z}}) (\mathbf{v}_1^H \mathbf{z}_n - \mathbf{v}_1^H \bar{\mathbf{z}})^H = \mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1, \quad (1.1)$$

where $\mathbf{S}_{ZZ}$ is the covariance matrix of $\mathbf{Z}$ given by

$$\mathbf{S}_{ZZ} = \frac{1}{N-1} \sum_{n=1}^N \mathbf{z}_n \mathbf{z}_n^H = \frac{1}{N-1} \mathbf{Z}^H \mathbf{Z}. \quad (1.2)$$

The goal is to find the vector $\mathbf{v}_1$ that maimizes this variance, formulated as:

$$\underset{\mathbf{v}_1}{\operatorname{argmax}} \quad \mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1 \quad \text{subject to} \quad \mathbf{v}_1^H \mathbf{v}_1 = 1. \quad (1.3)$$

To solve this optimization problem, we maximize the projected variance $\mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1$ with respect to $\mathbf{v}_1$. To enforce the normalization constraint $\mathbf{v}_1^H \mathbf{v}_1 = 1$, a Lagrange multiplier λ_1 is introduced, leading to the unconstrained maximization of:

³This normalization is for convenience and not necessary, since we are only interested in the direction of $\mathbf{v}_1$, not its magnitude.

$$\mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1 + \lambda_1 (1 - \mathbf{v}_1^H \mathbf{v}_1). \quad (1.4) \quad 1$$

By setting the derivative with respect to $\mathbf{v}_1$ to zero, we find a stationary point when: 2

$$\mathbf{S}_{ZZ} \mathbf{v}_1 = \lambda_1 \mathbf{v}_1. \quad (1.5) \quad 3$$

This indicates that $\mathbf{v}_1$ must be a eigenvector of $\mathbf{S}_{ZZ}$, with λ_1 being the corresponding real-valued⁴ eigenvalue. The variance $\mathbf{v}_1^T \mathbf{S}_{ZZ} \mathbf{v}_1$ reaches its maximum when $\mathbf{v}_1$ aligns with the eigenvector associated with the largest eigenvalue⁵ λ_1 . The eigenvector is referred to as the first **vector of coefficients**⁶ (VC) of the data set. 4
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To find additional VCs, we iteratively select new directions that maximize the projected variance while being orthogonal to the previously chosen directions. For an M-dimensional projection space, the optimal linear projection that maximizes the variance of the projected data is defined by the M eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_M$ of the covariance matrix $\mathbf{S}_{ZZ}$, corresponding to the M largest eigenvalues $\lambda_1, \dots, \lambda_M$. This result can be derived using proof by induction, leading to the concise expression: 8
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$$\mathbf{S}_{ZZ} \mathbf{V} = \mathbf{V} \mathbf{\Lambda}, \quad (1.6) \quad 14$$

where $\mathbf{V}$ is unitary, i.e. $\mathbf{V}^T \mathbf{V} = \mathbf{I}_M$. 15

Thus, the eigenvectors of $\mathbf{S}_{ZZ}$ emerge as the new coordiante system "along" regularities in the data. For cases where these regularities are not aligned in unitary directions, see Section 1.1.2. 16
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According to the spectral theorem for normal matrices [e.g. 55, p. 133], the hermitian covariance matrix $\mathbf{S}_{ZZ}$ can be decomposed into the sum of the outer products of the eigenvectors $\mathbf{v}_i$ and the corresponding real-valued eigenvalues λ_i : 18
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⁴This follows from the property of the covariance matrix being hermitian.

⁵This can easily verified by left-multiplying Equation (1.5) with $\mathbf{v}_1^H$ and making use of the normalization condition.

⁶Naming conventions can vary significantly across different scientific domains. For example, in machine learning, the terms principal components and principal component scores refer to the eigenvectors (here $\mathbf{V}$) and their corresponding data projections (here $\mathbf{P}$) [e.g. 6]. In statistics, these are often called vectors of coefficients/loadings and principal components [e.g. 64], while in climate science and meteorology, they are known as empirical orthogonal functions and principal components/expansion coefficients [e.g. 109]. In this thesis, I aim to maintain consistency with the terminology commonly used in statistics.

$$\mathbf{S}_{ZZ} = \sum_{i=1}^M \lambda_i \mathbf{v}_i \mathbf{v}_i^H. \quad (1.7)$$

Each eigenvalue λ_i represents the amount of variance captured by the corresponding component, typically expressed as a fraction γ_i of the total variance:

$$\gamma_i = \frac{\lambda_i}{\sum_{i=1}^M \lambda_i} \quad (1.8)$$

Projecting the data $\mathbf{Z}$ onto the M -dimensional subspace thus provides a lower dimensional representation, called the principal components (PCs),

$$\mathbf{P} = \mathbf{Z}\mathbf{V}, \quad (1.9)$$

which also enables the reconstruction of the original data from the projected data:

$$\mathbf{Z} = \mathbf{P}\mathbf{V}^H. \quad (1.10)$$

However, calculating the eigenvector decomposition is not always computationally efficient. A more efficient method involves using Singular Value Domposition (SVD) [e.g. 55, p. 150], which decomposes the data matrix as

$$\mathbf{Z} = \mathbf{U}'\mathbf{\Sigma}\mathbf{V}'^H, \quad (1.11)$$

where $\mathbf{U}'$ and $\mathbf{V}'$ are unitary matrices, $\mathbf{U}'^H\mathbf{U}' = \mathbf{V}'^H\mathbf{V}' = \mathbf{I}_M$, and $\mathbf{\Sigma}$ is a diagonal matrix containing the singular values σ_i of $\mathbf{Z}$. The covariance matrix $\mathbf{S}_{ZZ}$ can then be expressed in terms of the SVD as

$$\mathbf{S} = \frac{1}{N-1} \mathbf{Z}^H \mathbf{Z} = \frac{1}{N-1} \mathbf{V}'^H \mathbf{\Sigma}^H \mathbf{U}'^H \mathbf{U}' \mathbf{\Sigma} \mathbf{V}' = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^H, \quad (1.12)$$

where, by comparison with (1.6), $\mathbf{V} = \mathbf{V}'$ and $\mathbf{\Lambda} = \frac{1}{N-1} \mathbf{\Sigma}^T \mathbf{\Sigma}$. This establishes a direct relationship between the eigenvalues of the covariance matrix and the singular values of the data matrix:

$$\lambda_i = \frac{\sigma_i^2}{N-1}. \quad (1.13) \quad 1$$

Comparing (1.10) with (1.11) shows that the PC scores $\mathbf{P}$ can be normalized 2

$$\mathbf{U} \equiv \mathbf{U}' = \mathbf{P}\mathbf{\Sigma}^{-1} = \frac{1}{(N-1)^{1/2}} \mathbf{P}\mathbf{\Lambda}^{-1/2}, \quad 3$$

$$\text{or alternatively } \mathbf{P} = (N-1)^{1/2} \mathbf{U}\mathbf{\Lambda}^{1/2}. \quad (1.14) \quad 4$$

Thus, $\mathbf{U}^H \mathbf{U} = \mathbf{I}_M$ ensures that the PC scores $\mathbf{U}$ are unitary and uncorrelated. 5

A common scaling method for the vector of coefficients $\mathbf{V}$ involves scaling by the square root of the corresponding eigenvalues, referred to as **loadings**: 6

$$\tilde{\mathbf{V}} = \mathbf{V}\mathbf{\Lambda}^{1/2}. \quad (1.15) \quad 8$$

Although not strictly unitary, these loadings form a generalized unitary matrix, as shown by 9

$$\tilde{\mathbf{V}}^H \tilde{\mathbf{V}} = \mathbf{\Lambda}. \quad (1.16) \quad 10$$

Using this scaling, the Equation 1.10 can be expressed as 11

$$\mathbf{X} = (N-1)^{1/2} \mathbf{U}\tilde{\mathbf{V}}^T. \quad (1.17) \quad 12$$

Computational Considerations The computational cost of performing eigendecomposition on the covariance matrix scales with $O(D^3)$, making it impractical for modern datasets, such as those in climate science, where typical data sizes are on the order of $D = O(10^6)$ or even larger. For example, ERA5 [16], a widely used reference for state-of-the-art reanalysis climate data, distributes a single climate variable on a regular 0.25° grid, resulting in $D = 1\,036\,800$ grid cells. 13

In practice, more efficient algorithms, such as SVD or power methods [41], have been employed traditionally. Recently, randomized low-rank approximation techniques [29, 45] have further reduced computational complexity to $O(ND \log(M) + (N+D)M^2)$. These advancements offer feasible alternatives for handling large, out-of-memory datasets (see Table 1.1). 14
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Shortcomings Despite its widespread use and versatility, PCA has several limitations and potential drawbacks that should be considered:

1. **Linearity Assumption:** PCA assumes a linear structure in the data, which may not hold in complex systems such as climate data, where nonlinear relationships can be significant [68, 106].
2. **Interpretability of VCs:** The orthogonality of the vectors of coefficients is a mathematical construct that may not always correspond to the physical reality of the data, potentially leading to misinterpretations [100].
3. **Variance Focus:** Focusing on directions of maximum variance may not always capture the most relevant features for a specific application, potentially critical but less variable information [36].
4. **Static Nature:** As a static method, PCA does not account for temporal dynamics or changes over time, which is particularly limiting in climate science, where time evolution is a key aspect.
5. **Sensitivity to Outliers:** PCA can be sensitive to outliers, as extreme values can disproportionately influence the VCs, leading to misleading results [19].
6. **Sensitivity to Scaling:** The results are sensitive to the scaling of the input data, meaning that variables with larger ranges can dominate the VCs unless the data is appropriately standardized.
7. **Loss of Information:** While dimensionality reduction is often the goal, PCA inherently discards some information, which could lead to the loss of important details crucial for accurate analysis and interpretation in specific applications.

Over the past few decades, numerous modifications and extensions have been proposed to address these limitations, especially in the context of climate science. For instance, to address the linearity assumption, various forms of kernel PCA [13, 106, 121] have been developed. For improved interpretability, rotated [100] and sparse [65] PCA methods have been proposed. Robust PCA [19] offers a more resilient approach to handling sensitivity to outlier. Furthermore, extended [11, 120] and complex [3, 53] PCA methods have been introduced to account for temporal dynamics. In addition, eigenmethods like Optimal Persistence Analysis (OPA) [26, 27] and Principal Oscillation Pattern (POP) analysis [51] provide alternative approaches independent to variance maximization to understanding dominant patterns in physical climate fields.

In the following sections, we will explore some of these shortcomings and proposed solutions in greater detail. Specifically, we will discuss how additional rotation of the PCA solution can

Table 1.1 — Computational costs of different algorithms used to compute either full or approximative M -rank solutions for PCA for a real, dense, in-memory matrix with dimensions $N \times D$. Compiled from Bishop [6] and Halko *et al.* [45].

Algorithm	Complexity
Eigenvector decomposition	$O(D^3)$
Full SVD	$O(ND \min(N, D))$
Power method	$O(MD^2)$
Truncated SVD	$O(NDM)$
Randomized SVD	$O(ND \log(M) + (N + D)M^2)$

enhance physical interpretability (Section 1.1.2) and how complexifying the data can provide insights into time evolution patterns (Section 1.1.3).

Weighting of Data Variables In some cases, it may be necessary to weight the data variables differently. This can be achieved by scaling the data matrix $\mathbf{Z}$ by a diagonal matrix $\mathbf{W}$, where the j^{th} diagonal element w_j corresponds to the weight of the j^{th} variable. A very common situation is that the observed variables exhibit stark differences in their variance, which can sometimes lead to mask structure in the data as the PCA solution is solely driven by the different scales of variance (so the first principal axis will closely align to the variable with the highest variance) in which case the obvious choice is to use the inverse of the standard deviation of each variable, $w_j = 1/\sigma_j$, which ensures that all variables are equally weighted in which case the covariance matrix becomes the correlation matrix. Another common situation in climate science is that the physical fields are represented on regular longitude-latitude grids. This representation leads to a higher density of grid points near the poles, which typically results in an exaggerated importance of components spatially focused near the poles. To address this, the data matrix can be weighted by the square root of the cosine of the latitude [87], $w_j = \cos(\text{lat}_j)$, where lat_j is the latitude of the j^{th} grid point. This weighting ensures that the grid points are equally weighted in the PCA solution, regardless of their geographical location. In the following we will discuss some popular extensions of PCA together with a related multi-set method called Maximum Covariance Analysis (see Section 1.2), for which the latitude weighting can be applied in the same way.

1.1.2 Sparse PCA via Rotation

PCA is widely used for dimensionality reduction, as it maximizes the explained variance and provides orthogonality, making it optimal in a least-squares sense [6]. It effectively identifies principal axes that capture the covariance among variables, allowing for the examination of interrelationships between stations, grid points or variables. However, PCA is not ideal for identifying the cause of variation or deriving physical modes of variability [64, 74, 100].

One limitation is that real-world processes do not necessarily adhere to the orthogonality imposed by PCA's eigenvectors. Additionally, the dense nature of these VCs – where all elements are non-zero – can make interpretation challenging. In the context of climate and environmental science, where variables are often represented across spatial grids, these limitations can lead to recognizable but artificial patterns known as Buell patterns [12, 73, 100]. These patterns typically start with a positive component and evolve into increasingly complex structures, such as bipole or tripole patterns (Figure 1.1). Such issues highlight the disconnect between standard PCA and underlying physical processes [100], prompting the development of alternative methods that produce sparse VCs, characterized by many zero or near-zero elements. This sparsity – which necessarily implies the loss of orthogonality for at least one of the VCs and PC scores – can enhance interpretability by localizing the patterns and reducing the density of the VCs.

In climate science, the most common approach to achieving sparse PCA is through rotation techniques [54, 99, 100], a concept borrowed from factor analysis [cf. 64, pp. 269]. In this "factor analysis viewpoint" of PCA [64, pp. 150-161], a linear transformation is applied to a subset of the VCs to derive a new set of rotated VCs that are "simpler" in a heuristically defined manner, ideally aligning better with the underlying physical processes.

As noted in Equation (1.17), once the orthogonality constraint is relaxed, PCA solutions exhibit inherent indeterminacy, allowing any non-singular transformation $\mathbf{R}$ to satisfy the PCA formulation [e.g. 100, p. 305]:

$$\mathbf{Z} = (\mathbf{N} - 1)^{1/2} \mathbf{U} \tilde{\mathbf{V}}^H = (\mathbf{N} - 1)^{1/2} \mathbf{U} \mathbf{R} \mathbf{R}^{-1} \tilde{\mathbf{V}}^H. \quad (1.18)$$

Rotated PCA exploits this indeterminacy by choosing $\mathbf{R}$ to optimize a simplicity criterion f , resulting in rotated loadings⁷ $\tilde{\mathbf{V}}_R = \tilde{\mathbf{V}} \mathbf{R}$ that maximize f :

$$\underset{\mathbf{R}}{\operatorname{argmax}} f(\tilde{\mathbf{V}}_R). \quad (1.19)$$

⁷In principle, it is possible to rotate the unitary matrix of coefficients $\mathbf{V}$ instead of the loadings matrix $\tilde{\mathbf{V}}$ [64, 82]. However, this approach is seldom used in practice. The primary reason is that rotating the unitary coefficient matrix preserves the orthogonality of the VCs but results in the loss of orthogonality among the PCs, which does not address issues like the Buell patterns. Moreover, this method does not consider the relative importance of the spatial patterns, as indicated by the eigenvalues, making the results of the rotation process highly sensitive to the selection of the number of components to rotate.

There are two main categories of rotation: *orthogonal*⁸ and *oblique*. In orthogonal rotation, $\mathbf{R}$ is a unitary matrix, preserving the uncorrelated nature of the PCs since $\mathbf{U}_R^H \mathbf{U}_R = \mathbf{R}^H \mathbf{U}^H \mathbf{U} \mathbf{R} = \mathbf{I}_M$. In oblique rotation, $\mathbf{R}$ is a general non-singular matrix, resulting in correlated PCs, as $\mathbf{U}_R^H \mathbf{U}_R = \mathbf{R}^H \mathbf{R}$. Regardless of the method, the rotated loadings are not orthogonal,

$$\tilde{\mathbf{V}}_R^H \tilde{\mathbf{V}}_R = \mathbf{R}^H \tilde{\mathbf{V}}^H \tilde{\mathbf{V}} \mathbf{R} \stackrel{(1.16)}{=} \mathbf{R}^H \mathbf{A} \mathbf{R} \neq \mathbf{I}_M, \quad (1.20)$$

which is crucial for achieving a more interpretable solution. Various simplicity criteria f have been proposed [see e.g. 100], with most focusing on making the rotated loadings more localized and physically interpretable. In practice, the specific choice of simplicity criterion often has little impact on the results [100].

Varimax Orthogonal Rotation The most widely used orthogonal rotation technique is the varimax criterion [66], which aims to simplify the PCs by maximizing the variance of the squared loadings,

$$f(\tilde{\mathbf{V}}_R) = \sum_{m=1}^M \left(p \sum_{j=1}^P b_{jm}^2 - \left(\sum_{j=1}^P b_{jm} \right)^2 \right) \quad (1.21)$$

where b_{jm} are the squared norm of the rotated loadings $|\tilde{v}_{R,jm}|^2/h_j^2$ normalized by the communalities $h_j^2 = \sum_{m=1}^M |\tilde{v}_{R,jm}|^2$ of variable j . The varimax rotation pushes the elements of $\tilde{\mathbf{V}}_R$ towards either 0 or 1, creating a sparse solution with primarily large or small values, which is particularly useful for identifying localized physical processes, such as atmospheric or oceanic teleconnections.

Promax Oblique Rotation The promax criterion [52], a generalization of the varimax criterion, is one of the most common oblique rotation techniques. It is based on the premise that an orthogonal simple structure solution, such as varimax, is often close to an optimal oblique solution. The promax method constructs a hypothetical oblique target matrix $\mathbf{T}$ by performing an oblique Procrustes transformation [61] on the varimax-rotated loadings matrix,

⁸In the broader context of complex data, this would be referred to as a unitary rotation. However, since factor analysis has primarily been applied to real-valued data, the terms orthogonal and oblique rotations have become the standard. To avoid confusion, I use these historical terms while acknowledging that they also encompass the more general case of complex data.

$$\mathbf{T} = \tilde{\mathbf{V}}_{\mathbf{R}} \mathbf{R} + \mathbf{E}, \quad (1.22)$$

where $\mathbf{R}$ is the transformation matrix and $\mathbf{E}$ is a matrix of residuals. The normal equation for solving the complex Procrustes problem is given by:

$$\mathbf{R} = (\tilde{\mathbf{V}}_{\mathbf{R}}^{\mathbf{H}} \tilde{\mathbf{V}}_{\mathbf{R}})^{-1} \mathbf{T}^{\mathbf{H}} \tilde{\mathbf{V}}_{\mathbf{R}}. \quad (1.23)$$

To define the target matrix $\mathbf{T}$, the promax criterion increases the discrimination between high and low loadings by raising the loadings to a power q while retaining the original sign,

$$t_{jm} = \frac{|a_{jm}|^{q+1}}{a_{jm}}, \quad (1.24)$$

where q controls the degree of obliqueness and a_{jm} are the elements of the max-normalized, varimax-rotated loadings matrix $\mathbf{A}$, defined as

$$\mathbf{A} = \tilde{\mathbf{A}} \mathbf{K}^{-1}, \quad (1.25)$$

$$\tilde{\mathbf{A}} = \mathbf{H}^{-1} \tilde{\mathbf{V}}_{\mathbf{R}}, \quad (1.26)$$

with $\mathbf{K}$ being a diagonal matrix containing the maximum absolute values of each column of $\tilde{\mathbf{A}}$, $a_m = \max_m |A_{pm}|$, and $\mathbf{H}$ a diagonal matrix with the square root of the communalities h_j for the j^{th} variable.

Application and Criticism Both orthogonal and oblique rotation techniques, particularly varimax and promax, have been extensively used since the 1980s to identify stationary modes of variability in climate data [e.g. 4, 40, 54, 62, 72, 77, 83]. Notable climate oscillations, such as the North Atlantic Oscillation (NAO), have been defined based on varimax-rotated patterns of loadings [4]. These techniques also help identify "ghost patterns", artificial patterns identified by PCA that are not physically meaningful [62, 76].

Rotated modes offer several advantages, such as greater robustness against temporal and spatial subsampling [22] compared to unrotated modes. They also effectively mitigate the statistical artifacts associated with Buell patterns, resulting in more realistic representations of the underlying autocorrelation structure and thus increased interpretability [76, 100, 101].

Moreover, rotation techniques have been successfully interated into other PCa extensions dealing with the non-linear and non-stationary nature of climate oscillations [13, 44, 53].

However, rotation techniques are not without criticism, mainly due to the ad-hoc nature of the various simplicity criteria [cf. 64, pp. 270-274]. This has led to the exploration of alternative approaches to enforce sparsity. Notably, Jolliffe *et al.* [65] introduced the Simplified Component Technique - LASSO (SCoTLASS), a sparse PCA variant based on the least absolute shrinkage and selection operator (LASSO), originally from regularized regression [112]. This method imposes a constraint on the ℓ_1 norm of the vector of loadings, resulting in some loading coefficients being exactly zero.

Despite its advantages, SCoTLASS has drawbacks, particularly when the number of variables D is much larger than the number of observations N . In such cases, SCoTLASS can only select at most N non-zero elements in the loading vectors, which is not ideal in climate science where often $D \gg N$. Additionally, since SCoTLASS solutions cannot be obtained via SVD and require a numerical approach using projected gradient methods [114], the computational demand is typically higher than for (rotated) PCA. While this method has inspired many other sparse PCA techniques based on penalties or constraints (usually ℓ_0 or ℓ_1) [see e.g. 34, 125, 126], these formulations are often complex and require algorithmic innovations to avoid issues like local optima and ensure convergence. They may also face challenges with memory limitations, slow computation speeds and limited effectiveness in modelling the data [18, 43].

It is likely for these reasons, that rotated PCA remains one of the most commonly used techniques for achieving sparse PCA solutions in climate science. It is computationally efficient, straightforward to implement and provides a more interpretable understanding of the data compared to standard PCA.

1.1.3 Hilbert PCA

While PCA efficiently identifies the principal axes along which the covariance among all variables is distributed, and rotated PCA provides a more sparse solution related to underlying physical processes, both methods are inherently static. They do not for temporal dynamics or changes over time, as they only consider the current state of the physical field. This limitation is significant in climate science, where time evolution is crucial. For instance, standard PCA does not differentiate between the order of data points over time, resulting in the same eigenvectors and eigenvalues regardless of temporal permutations.

This static nature poses a challenge for capturing oscillatory patterns, such as QBO, MJO, ENSO and BSISO, which involve spatial patterns propagating over time. These phenomena cannot be adequately captured by standard PCA.

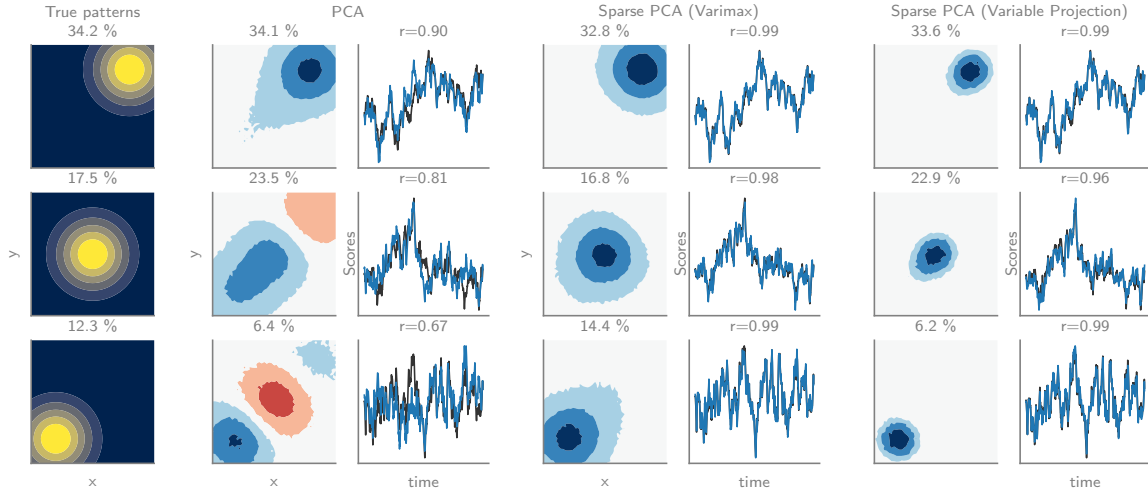


Figure 1.1 — Comparison of the first three modes obtained from PCA vs. Varimax-rotated PCA for a synthetic data set. (Left) Original data set with three different, uncorrelated autoregressive processes. (Middle) PCA solution. (Right) Varimax-rotated PCA solution.

1 Hilbert Transform as a Momentum Operator

To address this limitation, one approach is to model both the current state of the climate and its tendency or ‘momentum’, analogous to the Hamiltonian principle in mechanics, where a system’s future is described by first-order differential equations for location (state) and momentum (tendency) [cf. 109, p. 353]. In classical mechanics, propagating features are represented using canonical coordinates: each state variable is augmented by a momentum variable, allowing oscillations to be depicted as rotations in a two-dimensional phase space. From a data perspective, this augmentation can be achieved via the Hilbert transform ($\mathcal{H}$). For a real time series $x(t)$, the Hilbert transform $x'(t)$ is defined as

$$x'(t) \equiv \mathcal{H}\{x\}(t) = \frac{1}{\pi} \text{p.v.} \int_{-\infty}^{\infty} \frac{x(\tau)}{\tau - t} d\tau, \quad (1.27)$$

where p.v. denotes the Cauchy principal value ensuring convergence despite the singularity at $\tau = t$. The Hilbert transform emphasises the local properties of $x(t)$ by convolution with $\frac{1}{\pi t}$. In the frequency domain, it is related to the Fourier transform $\mathcal{F}$ by

$$\mathcal{F}\{y\}(\omega) = -i \operatorname{sgn}(\omega) \mathcal{F}\{x\}(\omega) \quad (1.28)$$

where the imaginary unit i indicates a phase shift of $\pi/2$ for the frequency components ω of x , placing x' in quadrature to x . This provides an intuitive interpretation of the Hilbert transform, as it acts as a linear filter that alters the phase of the signal's components by 90 degrees without changing their amplitude.

By using the Hilbert transform, a real-valued time series $x(t)$ can be complexified into $z(t) = x(t) + i\mathcal{H}\{x\}(t)$, known as the *analytic signal*, which allows for the computation of the instantaneous amplitude and phase of $x(t)$:

$$A(t) = |z(t)| = \sqrt{x(t)^2 + \mathcal{H}\{x\}(t)^2}, \quad \phi(t) = \arg(z(t)) = \arctan\left(\frac{\mathcal{H}\{x\}(t)}{x(t)}\right). \quad (1.29)$$

Similar results can be obtained for the discrete case [89, pp. 942-956], and this concept can be extended to multivariate data [e.g. 46, pp. 105-106], defining the analytic signal of a data matrix X as

$$Z = X + i\mathcal{H}\{X\}, \quad (1.30)$$

where $\mathcal{H}\{X\}$ is the feature-wise Hilbert transform of X .

Rasmusson *et al.* [97] introduced the concept of Hilbert PCA using this data augmentation technique⁹. This method, refined by Barnett [3] and Horel [53], involves applying PCA to the augmented data matrix Z . By incorporating phase information, Hilbert PCA can capture the direction and speed of wave-like patterns, unlike traditional PCA, which only identifies standing oscillations. The resulting eigenvectors and principal component scores are complex, with the real and imaginary components linked via the Hilbert transform. The variance capture by each component is represented by the real eigenvalues of the complex covariance matrix of Z (see Section 1.1).

The main advantage of Hilbert PCA is the ability to identify and reconstruct propagating waves, which is especially valuable in studying climate phenomena characterized by such patterns, including MJO, ENSO, QBO and BSISO. For instance, Cai & Baines [17] utilized Hilbert PCA to analyze SST in the southern hemisphere, linking the findings to ENSO, while Bürger [15] employed it to develop an index for BSISO. However, to reliably identify oscillatory patterns, sufficiently long time series are required to resolve at least one full oscillation [81]. Additionally, interpreting of the Hilbert transform as representing the instantaneous frequency is valid holds for narrow-band signals; the phase of the Hilbert transform is not well-defined

⁹Although Rasmusson *et al.* [97] originally termed it *Hilbert Singular Decomposition*, the terms *Hilbert EOF/PCA* and *Complex EOF analysis* are more commonly used in the literature [15, 46, 64, 109].

for broadband signals [8, 39]. Moreover, real-time monitoring is challenging with Hilbert PCA because new real-valued data cannot be directly projected onto the existing patterns without knowing or estimating the exact momentum (phase) of the new data, which requires future information (Equation (1.27)). To mitigate this, an approximative real-time (causal) Hilbert transform can be employed [14, 15], offering a practical remedy to the issue of real-time applicability. Finally, Hilbert PCA shares PCA's limitations regarding physical interpretability due to the orthogonality of the VCs. To address this, rotated Hilbert PCA has been proposed to enhance interpretability [7, 54].

1.2 Maximum Covariance Analysis

In contrast to finding latent variables that maximize covariance within a single dataset, Maximum Covariance Analysis (MCA) aims to identify the latent variables that are common in two sets of data and maximize covariance. For example, in the case of the MJO [80], which is the dominant mode of intraseasonal variability in the tropical atmosphere, characterized by an eastward-moving cell of overturning wind circulation and associated regions of enhanced and suppressed precipitation, MCA helps in findings shared dynamics between different variables. This method allows us to focus on shared variability while ignoring dominant variations associated with only one variable, such as diurnal cycles linked to surface processes absent at higher atmosphere levels.

MCA identifies patterns of maximum covariance between datasets by finding pairs of linear subspaces with high cross-covariance [e.g. 109, p. 321]. Unlike PCA, which operates on a single set of variables, MCA considers two¹⁰ sets of variables and seeks components that exhibit high covariance across datasets. This approach can be formulated as an optimization problem, aiming to find pairs of (complex-valued) coupled patterns $\mathbf{v}_{s,1}$ for datasets $\mathbf{Z}_s$, $s = x, y$, such that the projection maximizes the covariance between the datasets. The solution involves a generalized eigenvector problem, often solved using the Singular Value Decomposition (SVD) of the cross-covariance matrix.

$$\operatorname{argmax}_{\mathbf{v}_{x,1}, \mathbf{v}_{y,1}} \mathbf{v}_{x,1}^T \mathbf{C}_{xy} \mathbf{v}_{y,1} \quad \text{subject to} \quad \|\mathbf{v}_{x,1}\|_2 = \|\mathbf{v}_{y,1}\|_2 = 1, \quad (1.31)$$

where the cross-covariance matrix between the two datasets is given by

¹⁰In theory, this can be extended to more than two sets of data [67].

$$\mathbf{C}_{xy} = \frac{1}{N-1} \mathbf{Z}_x^H \mathbf{Z}_y. \quad (1.32) \quad 1$$

The solution to this problem gives rise to a generalized eigenvalue equations similar to that of PCA, which can also be solved by using the SVD of the cross-covariance matrix 2
3

$$\mathbf{C}_{xy} = \mathbf{V}_x \mathbf{\Sigma} \mathbf{V}_y^H \quad (1.33) \quad 4$$

where the i^{th} column of the left and right singular vectors, $\mathbf{V}_x$ and $\mathbf{V}_y$, contains the pattern vectors $\mathbf{v}_{x,i}$ and $\mathbf{v}_{y,i}$, and $\mathbf{\Sigma}$ a diagonal matrix containing the singular values σ_i in decending order. The first left and right singular vectors correspond to the maximum covariance, given by the largest singular value σ_1 . 5
6
7
8

Since the pattern vectors are unitary, datasets can be expanded as 9

$$\mathbf{Z}_s = \mathbf{P}_s \mathbf{V}_s^H, \quad (1.34) \quad 10$$

with the expansion coefficients $\mathbf{P}_s$. The expansion coefficients $\mathbf{P}_x$ and $\mathbf{P}_y$ are uncorrelated with each other: 11
12

$$\mathbf{P}_x^H \mathbf{P}_y = \mathbf{V}_x^H \mathbf{Z}_x^H \mathbf{Z}_y \mathbf{V}_y \stackrel{(1.32)}{=} (N-1) \mathbf{V}_x^H \mathbf{C}_{xy} \mathbf{V}_y \stackrel{(1.33)}{=} (N-1) \mathbf{\Sigma}. \quad (1.35) \quad 13$$

However, the expansion coefficients $\mathbf{P}_x$ and $\mathbf{P}_y$ are generally correlated within themselves as indicated by the condition $\mathbf{P}_s^H \mathbf{P}_s \neq c \mathbf{I}_M$ for any arbitrary constant c . 14
15

The total covariance, given by the squared Frobenius norm of the cross-covariance matrix, is 16

$$\|\mathbf{C}_{xy}\|_F^2 = \sum_{i=1}^M \sigma_i^2, \quad (1.36) \quad 17$$

which shows that each mode captures a fraction of the total squared covariance. The relative importance of each mode can be quantified by the squared covariance fraction $f_{xy,i}$: 18
19

$$f_{xy,i} = \frac{\sigma_i^2}{\sum_{i=1}^M \sigma_i^2}. \quad (1.37)$$

This fraction provides a measure of how much of the shared variability between the two datasets is captured by the i^{th} mode. To assess the importance of the patterns themselves, it is necessary to consider the variance within the individual datasets explained by each mode relative to the total variance of the individual datasets. This can be expressed using the squared Frobenius norm of the individual datasets:

$$f_{xx,i} = \frac{\mathbf{v}_{x,i}^H \mathbf{C}_{xx} \mathbf{v}_{x,i}}{\|\mathbf{Z}_x\|_F^2} = \frac{1}{N-1} \frac{\mathbf{p}_{x,i}^H \mathbf{p}_{x,i}}{\|\mathbf{Z}_x\|_F^2}, \quad (1.38)$$

$$f_{yy,i} = \frac{\mathbf{v}_{y,i}^H \mathbf{C}_{yy} \mathbf{v}_{y,i}}{\|\mathbf{Z}_y\|_F^2} = \frac{1}{N-1} \frac{\mathbf{p}_{y,i}^H \mathbf{p}_{y,i}}{\|\mathbf{Z}_y\|_F^2}. \quad (1.39)$$

While the squared covariance fraction $f_{xy,i}$ indicates the amount of shared variability between the two datasets, it does not provide information about the strength of the association or coupling between them. To measure this, we use the correlation coefficient ρ_i for the expansion coefficients of mode i between the two datasets:

$$\rho_i = \frac{\text{cov}(\mathbf{p}_{1,i}, \mathbf{p}_{2,i})}{(\text{var}(\mathbf{p}_{1,i}) \text{var}(\mathbf{p}_{2,i}))^{1/2}} = \frac{\sigma_i}{\|\mathbf{p}_{1,i}\|^2 \|\mathbf{p}_{2,i}\|^2}. \quad (1.40)$$

The homogeneous patterns [10] for the pattern vectors are:

$$\text{HoP}_{x,i} = \text{cov}(\mathbf{Z}_x, \mathbf{p}_{x,i}) = \mathbf{S}_{xx} \mathbf{v}_{x,i}, \quad (1.41)$$

$$\text{HoP}_{y,i} = \text{cov}(\mathbf{Z}_y, \mathbf{p}_{y,i}) = \mathbf{S}_{yy} \mathbf{v}_{y,i}, \quad (1.42)$$

These patterns provide an indicator of which variables or features are important for the covarying components in both fields. The corresponding heterogeneous patterns are defined as:

$$\text{HeP}_{x,i} = \text{cov}(\mathbf{Z}_x, \mathbf{p}_{y,i}) = \sigma_i \mathbf{v}_{y,i}, \quad (1.43) \quad 1$$

$$\text{HeP}_{y,i} = \text{cov}(\mathbf{Z}_y, \mathbf{p}_{x,i}) = \sigma_i \mathbf{v}_{x,i}. \quad (1.44) \quad 2$$

These patterns indicate how well the variables in one field can be predicted from the i^{th} expansion coefficient of the other field. One may also use standardized data, $\mathbf{Z}_x$ and $\mathbf{Z}_y$, to obtain the respective homogeneous and heterogeneous correlation patterns. 3
4
5

MCA gained popularity in the field of meteorology¹¹ through the works of Bretherton *et al.* [10] and Wallace *et al.* [118], since when it has been extensively applied [e.g. 5, 75, 86, 98, 115, 118]. 6
7

The method's orthogonal patterns ensure efficient dimensionality reduction of the covariance but limit physical interpretability, similar to PCA. To enhance interpretability, rotation of the MCA solution can be employed, as proposed by Cheng & Dunkerton [21], and later applied by Frankignoul & Kestenare [37]. A composite matrix of the left and right singular vectors is constructed and a simplicity criterion (e.g. varimax) is applied: 8
9
10
11
12

$$\mathbf{V}_R = \begin{bmatrix} \mathbf{V}_x \\ \mathbf{V}_y \end{bmatrix} \mathbf{M} \quad (1.45) \quad 13$$

In this case, $\mathbf{M}$ is a diagonal matrix that weights the augmented pattern matrix. While Cheng & Dunkerton [21] proposed $\mathbf{M} = \Sigma^{1/2}$, this choice is not ideal as the singular values σ_i are not directly related to the explained covariance of the patterns. Instead, it is more appropriate to weight each mode by the squared singular value, preserving the squared covariance fraction of the first m modes before and after (orthogonal) rotation. Thus, we use 14
15
16
17
18

$$\mathbf{M} = \Sigma \quad (1.46) \quad 19$$

for the weighted (or "loaded") pattern vectors. 20

Beyond rotation for better interpretability, Hilbert MCA can be applied to capture the temporal evolution of coupled patterns, akin to Hilbert PCA (see Section 1.1.3). This is especially useful for analyzing propagating patterns like the MJO, which involve covarying patterns in 21
22
23

¹¹According to Jolliffe [64, p. 225], the technique was first used in psychology in the 1950s under the name *inter-battery factor analysis*. As a variant of Partial Least Squares (PLS), it was later applied in biomedical sciences by Sampson *et al.* [103] and was referred to as *PLS-SVD* by Wegelin [122]. In meteorology, Bretherton *et al.* [10] introduced the term *SVD analysis*, which remains in use today [e.g. 86]. Occasionally, the method is also called *Canonical Covariance Analysis*, similar to Canonical Correlation Analysis [e.g. 23, 46].

different climate variables, such as wind speed in the troposphere and lower stratosphere, and precipitation. By augmenting data with the Hilbert transform, we obtain complexified fields and solve the maximization problem (Equation (1.31)) for these augmented data, capturing the temporal evolution of coupled patterns. It seems that Hilbert MCA has seen limited application in climate science [e.g. 32]. In this thesis we utilize the combined power of Hilbert and rotated MCA to identify and interpret coupled patterns of variability in the climate system.

PCA, MCA and CCA MCA reduces to PCA when both sets are identical. Furthermore, MCA can be viewed as two PCAs followed by rotation to align the results of the two analyses as closely as possible [24]. MCA is also closely related to Canonical Correlation Analysis (CCA), which, instead of maximizing shared covariance, seeks patterns that maximize correlation. Unlike MCA, where pattern vectors are by construction orthogonal, yielding correlated expansion coefficients, CCA results in pattern vectors that are not necessarily orthogonal but produce uncorrelated canonical variates [64, 123]. In fact, PCA, MCA and CCA can all be seen as special cases of a more general framework [9, 110].

1.3 Neural Networks

1.3.1 Introduction to Neural Networks

Definition and Overview: Explain what neural networks are, including a brief history and their importance in machine learning. **Basic Architecture:** Describe the basic structure of a neural network, including neurons, layers (input, hidden, output), activation functions, and the learning process.

1.3.2 Recurrent Neural Networks

Introduction to RNNs¹²: Describe the architecture of RNNs, their ability to handle sequential data, and their applications. **Challenges with RNNs:** Explain issues such as vanishing and exploding gradients.

Long Short-Term Memory (LSTM) Networks

Introduction to LSTM¹³: Discuss the motivation behind LSTM networks and how they address the limitations of standard RNNs. **LSTM Architecture:** Describe the components of LSTM cells (input gate, forget gate, output gate) and their functions. **Mathematical Formulation:** Provide

¹²Graves 2021 Supervised Sequence Labelling with Recurrent Neural Networks

¹³Hochreiter & Schmidhuber 1997

the equations governing LSTM networks. Applications of LSTM: Highlight key applications where LSTMs have shown significant performance improvements.

1.3.3 Explainable AI Techniques

Introduction to Explainable AI: Explain the need for explainable AI, particularly in critical applications where understanding model decisions is crucial. Overview of Techniques: Briefly list various explainable AI methods, with a focus on LRP and SHAP.

Layerwise Relevance Propagation (LRP)

Concept of LRP¹⁴: Describe what LRP is and its importance in understanding neural network decisions. Mechanism of LRP: Explain how LRP works, including the propagation rules and the relevance score computation. Applications of LRP: Provide examples of how LRP is used in practice to interpret model predictions.

simplest continuous latent variable model assumes Gaussian distributions for both the latent and observed variables. simplest continuous latent variable model assumes Gaussian distributions for both the latent and observed variables

SHapley Additive exPlanations (SHAP)

Introduction to SHAP¹⁵: Discuss the SHAP method, its foundation in cooperative game theory, and its advantages. SHAP Values: Explain how SHAP values are computed and their interpretation. Applications of SHAP: Provide examples of SHAP in various domains, highlighting its ability to provide consistent and locally accurate explanations.

1.4 Gaussian Processes

Class of pattern recognition techniques that typically require a metric to be defined that measures the similarity of any two vectors in input space, and are generally fast to "train" but slow at making predictions for test data points.

¹⁴Bach et al. 2015

¹⁵Lundberg 2017

1.4.1 Introduction to Gaussian Processes

Definition and Overview: Explain what Gaussian Processes are, their significance in machine learning, and their use in modeling continuous data. Basic Concepts: Introduce key concepts such as Gaussian distributions, covariance functions, and kernels.

1.4.2 Gaussian Processes for Regression

GP Regression Model: Describe the formulation of GP for regression tasks, including prior, likelihood, and posterior distributions. Kernel Functions: Discuss common kernel functions (e.g., squared exponential, Matérn) and their role in defining the GP's properties. Hyperparameter Tuning: Explain how to tune hyperparameters using techniques like maximum likelihood estimation.

1.4.3 Spatial Modeling with Gaussian Processes

Spatial Data: Define spatial data and the importance of modeling spatial dependencies. GPs in Spatial Modeling: Explain how GPs are particularly suited for spatial data due to their ability to model complex, non-linear relationships and provide uncertainty estimates.

Log Gaussian Cox Process

Introduction to Cox Processes: Explain Cox processes and their relevance in spatial point process modeling. Log Gaussian Cox Model: Describe the log Gaussian Cox model, which assumes the log intensity function is modeled by a GP. Mathematical Formulation: Provide the mathematical formulation of the log Gaussian Cox model.

Negative Binomial Distribution

Introduction to Negative Binomial Distribution: Explain the negative binomial distribution and its use in modeling overdispersed count data. Integration with Log Gaussian Cox Model: Describe how the negative binomial distribution can be integrated into the log Gaussian Cox model to handle count data with overdispersion.

1.4.4 Model Implementation and Inference

Model Fitting: Discuss methods for fitting the log Gaussian Cox model with a negative binomial distribution, including Markov Chain Monte Carlo (MCMC) and variational inference.

1.5 Comparative Perspective

21

Inference and Prediction: Explain how to perform inference and make predictions using the fitted model.

1

2

1.5 Comparative Perspective

3

Dimensionality Reduction Techniques in Climate Science: Bridging Theory and Practice

2

Over the past few decades, Earth system sciences have experienced an exponential increase in the volume and complexity of data [90], driven by the proliferation of Earth observations satellites [42], high-resolution climate models [49] and citizen science contributions [38]. As a result, researchers must now navigate the immense challenge of analyzing vast datasets to extract meaningful insights about Earth's climate system [104]. Dimensionality reduction techniques have become indispensable for addressing this challenge by simplifying complex data and revealing key underlying patterns.

In the previous chapter, we explored the theoretical foundations of several dimensionality reduction techniques commonly applied in Earth system sciences. We demonstrated how common PCA extensions can enhance MCA, yielding more physically interpretable patterns that reflect the shared dynamics between two datasets. This chapter we shift focus to the practical challenges that arise when applying these techniques to the increasingly large Earth system datasets and the solutions provided by open-source software. In particular, Section 2.3 will introduce the XEOFS Python library, which is designed to facilitate scalable dimensionality reduction with a particular focus on climate science. This package provides user-friendly access to both traditional and advanced methods, enabling researchers to conduct sophisticated analyses without needing extensive expertise in data science. This section is based upon the work presented in:

- **Rieger, N.** & Levang, S. J. xeoofs: Comprehensive EOF Analysis in Python with Xarray. *Journal of Open Source Software* 9, 6060. doi:[10.21105/joss.06060](https://doi.org/10.21105/joss.06060) (2024)

2.1 Open-Source Tools for Scalable Climate Data Analysis

Climate science is inherently data-intensive, relying on the analysis of large, complex datasets to understand the Earth's climate system. However, traditional analysis methods often struggle

24 Dimensionality Reduction Techniques in Climate Science: Bridging Theory and Practice

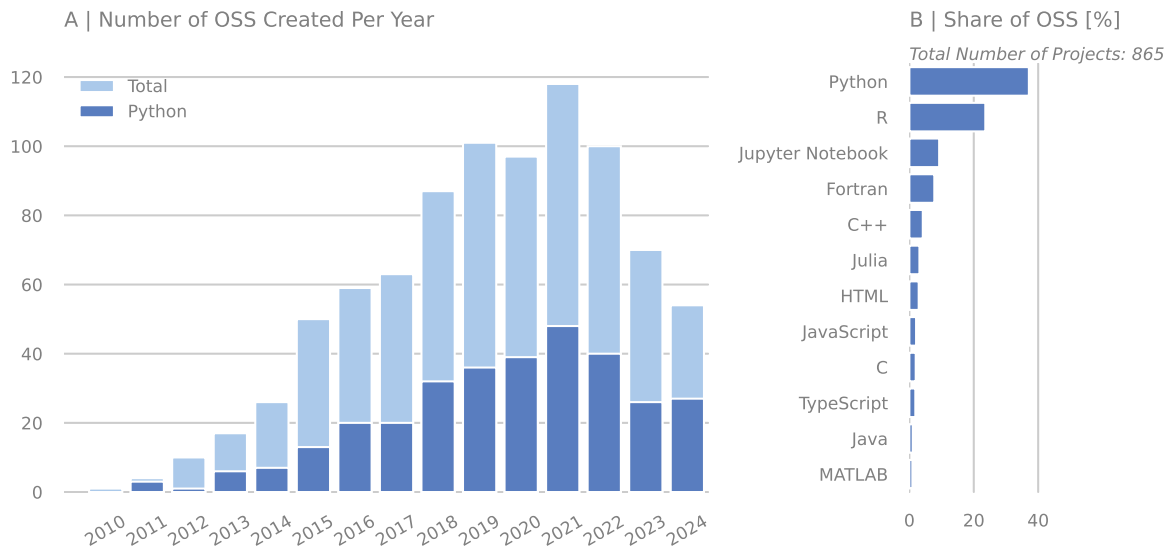


Figure 2.1 — Open-source software (OSS) projects in Earth system sciences. Panel A illustrates the number of new OSS projects created annually, highlighting the growing importance of Python-based projects. Panel B shows the relative share of programming languages used in all OSS projects since 2010, excluding languages with a share of $\leq 0.5\%$. The OSS project data is sourced from *Open Sustainable Technology* (OST) [1]. For this analysis, Earth system science projects include the OST categories *Atmosphere*, *Climate Change*, *Cryosphere*, *Hydrosphere* and *Biosphere*. Data for 2024 includes only the first half of the year, up to August.

to handle the scale and complexity of modern climate data, making the application of these techniques increasingly difficult [35]. This creates a growing need for tools that lower the barrier to entry for non-experts, allowing them to manage and analyze vast datasets effectively.

One of the central challenges in climate science today is the "data deluge" – the overwhelming volume of climate data and the difficulty of extracting useful insights from it. As Schnase *et al.* [105] noted, 'data analytics, the capacity to perform useful scientific analyses over large quantities of data in reasonable amounts of time' is a critical challenge for climate science. Without effective tools to transform large datasets into actionable knowledge, the mission of climate science falters.

To ensure that data analysis remains accessible to the broader scientific community, free, robust, and scalable software tools are essential [30]. These tools must allow researchers to work with large datasets, even without specialized programming skills, and simplify the processing of data produced by climate models.

It therefore comes as no surprise that the open-source software (OSS) movement has gained significant traction in the scientific community (Figure 2.1), with researchers increasingly recognizing its benefits for fostering collaboration, enhancing transparency, and ensuring reproducibility in research [63, 69, 85]. Python, in particular, has become a preferred language

for scientific computing (Figure 2.1 B), presumably due to its ease of use, extensive library support and active developer community. Notable examples of successful Python-based OSS projects include NUMPY [47, 88], SCIPY [88, 117] and MATPLOTLIB [59] libraries, which have become essential tools for scientific computing and visualization contributing to the democratization of scientific tools, driving innovation and facilitating new scientific discoveries [108].

The *Pangeo* project [91] has been especially influential in promoting open-source tools for big data geoscience. Two key components of the Pangeo ecosystem are the XARRAY [57] and DASK [25] Python libraries. XARRAY enhances the user experience by introducing labeled data structures that simplify the manipulation of multidimensional arrays, while DASK enables parallel and distributed computing. Together, these tools have revolutionized the way climate researchers handle large datasets, allowing them to focus on scientific questions rather than technical hurdles.

2.2 Challenges in Large-Scale and Multidimensional Climate Data Analysis

Analyzing climate data often involves multivariate statistical techniques to identify patterns within large, multidimensional datasets. In Section 1, we only explored a small subset of available methods for pattern identification in climate science [see e.g. 46, 58]. Most existing techniques assume that data can be structured as a two-dimensional (2D) matrix. However, climate data, whether from Earth observations or simulations, often exhibit two key characteristics:

1. **Multi-dimensionality:** Climate data span several dimensions, including spatial (longitude, latitude, height), temporal (time steps, forecast lead times), and categorical (variables, sensors).
2. **Large Volume:** These datasets are often so large that they exceed the memory capacity of a single computer and even of high performance computing (HPC) infrastructures.

These characteristics have led to the widespread adoption of tools like XARRAY and DASK which facilitate the handling of large, multidimensional datasets in the Python ecosystem.

However, applying dimensionality reduction techniques such as PCA and MCA (Section 1) presents several challenges. Climate datasets typically require significant pre-processing, including variable weighting (Section 1.1), removing invalid grid points (e.g., land cells in sea surface temperature data), and reshaping multidimensional arrays into 2D matrices. After analysis, researchers must reverse this process – transforming the 2D results back into

26 Dimensionality Reduction Techniques in Climate Science: Bridging Theory and Practice

1 their original multidimensional form, reinserting missing data (such as land grid cells), and
2 removing weights. This workflow can be cumbersome, error-prone, and time-consuming.

3 Earlier tools, such as the EOFs and XMCA libraries, addressed PCA and MCA applications in
4 climate science. However, they lack full compatibility with XARRAY and DASK limiting their
5 scalability and effectiveness when dealing with large datasets. For example, these libraries
6 do not support XARRAY's Dataset object, and neither leverages randomized linear algebra
7 techniques [33, 45], which can significantly improve computational performance for large
8 datasets.

9 In addition, researchers often need more than dimensionality reduction. Assessing the statisti-
10 cal significance of the extracted patterns, for example, requires bootstrapping techniques [31],
11 adding further complexity to the analysis process. Implementing these advanced methods
12 often requires substantial coding expertise, which can be a barrier for many scientists [124].

13 Moreover, several classical methods developed over the years for pattern extraction in climate
14 science have not yet been implemented in Python or integrated with XARRAY and DASK. This
15 lack of availability limits researchers' ability to apply these well-established techniques to their
16 datasets.

17 2.3 The XEOFS library: A Solution For Dimensionality Reduction in 18 Climate Science

19 The XEOFS library [102] was developed to address these challenges by providing a Python
20 package for applying dimensionality reduction techniques in climate science. Built with com-
21 patibility in mind, XEOFS integrates seamlessly with XARRAY and DASK, making it suitable for
22 large, multidimensional datasets. The library offers a user-friendly interface to both traditional
23 and advanced methods commonly used in climate data analysis (Table 2.1). Its development
24 adheres to key principles of open-source, scientific programming [2, 96], including clean coding
25 practices, rigorous test design [60], version control [94], robustness [111], usability [78], and
26 thorough documentation¹ [71]. The library also follows best practices in open-source software
27 development. For Python specifically, XEOFS aligns with the guidelines set forth by the Pangeo
28 project [91] and the *pyOpenSci* Packaging Guide [119].

29 Key features of the XEOFS library include:

30 **Multi-dimensional & Labeled Data** Designed specifically for XARRAY objects, XEOFS retains
31 data labels and dimensional structures throughout the analysis, which helps to maintain
32 clarity and to reduce errors.

¹The XEOFS online documentation is available at <https://xeofs.readthedocs.io>

Table 2.1 — Overview of available methods in the XEOFS library [102] (as of September 2024) for single-set (one dataset), cross-set (two datasets), and multi-set (more than two datasets) analysis. PCA: Principal Component Analysis; POP: Principal Oscillation Pattern; OPA: Optimal Persistence Analysis; CCA: Canonical Correlation Analysis; RDA: Redundancy Analysis; MCA: Maximum Covariance Analysis; CPCCA: Continuum Power CCA. Rotation techniques include orthogonal varimax [66] and oblique promax [52]. Combinations like rotated Hilbert PCA [53] are also supported. Multi-set CCA is based on the MCCA class of the CCA-ZOO library [20] and Sparse PCA uses the RISTRETTO library [33].

	METHOD	REFERENCE
SINGLE-SET ANALYSIS	PCA	[56, 79, 92]
	Complex PCA	[70, 109]
	Hilbert PCA	[3, 54, 97]
	Rotated PCA	[54, 99]
	Extended PCA	[11, 120]
	GWPCA	[48]
	Sparse PCA	[34]
	POP	[51]
	OPA	[26, 27]
CROSS-SET ANALYSIS	CCA	[10, 56]
	RDA	[109]
	MCA	[10, 109]
	CPCCA	[110]
	Complex CCA	
	Complex RDA	
	Complex MCA	
	Hilbert CCA	
	Hilbert RDA	
	Hilbert MCA	[32]
	Rotated MCA	[21]
	Rotated CPCCA	
MULTI-SET ANALYSIS	CCA	[67]
	Regularized CCA	[116]

28 Dimensionality Reduction Techniques in Climate Science: Bridging Theory and Practice

1 **Out-of-Memory Computations** Leveraging DASK, XEOFS can process large datasets that ex-
2 ceed the memory capacity of a single machine or HPC systems.

3 **Computational Efficiency** By employing randomized linear algebra, XEOFS dramatically im-
4 proves the efficiency of dimensionality reduction techniques, particularly for large
5 datasets. Benchmark comparisons show that XEOFS is approximately an order of magni-
6 tude faster than EOFs for medium to large datasets (Figure 2.2).

7 Additionally, XEOFS offers several auxiliary features:

8 **Handling Missing Values** The library can manage typical cases of missing data, such as
9 masked ocean or land cells.

10 **Weighted Analysis** XEOFS supports weighted analyses, allowing researchers to quickly apply
11 common weighting schemes (e.g. standard deviation or latitude-based cosine weighting,
12 see Section 1.1). It also supports custom weighting schemes.

13 **Significance Testing** A built-in bootstrapping interface enables users to test the significance
14 of extracted modes, reducing the need for additional coding.

15 **Model Serialization** Trained models can be saved and loaded in HDF5, netCDF4 or cloud-
16 optimized Zarr [84] formats ensuring compatibility with standard climate data formats
17 [50].

18 The XEOFS library is modular and object-oriented, allowing for future extensions and in-
19 tegration of new dimensionality reduction methods. It provides an intuitive application
20 programming interface (API) similar to the popular machine learning SCIKIT-LEARN library
21 [93] that simplifies the application of these techniques, even for researchers with limited
22 programming experience.

23 The library is available on the Python Package Index (PyPI) and can be installed via pip and
24 conda. Its source code is hosted on GitHub, where users can contribute to its development and
25 report any issues. Licensed under the MIT license, XEOFS is free to use and modify for both
26 academic and commercial purposes.

27 2.4 Conclusion

28 The development of the XEOFS library represents a contribution toward addressing the chal-
29 lenges of scalable and efficient dimensionality reduction in climate data analysis. However,
30 there are several opportunities for further refinement and expansion to better serve the needs
31 of the climate science research community.

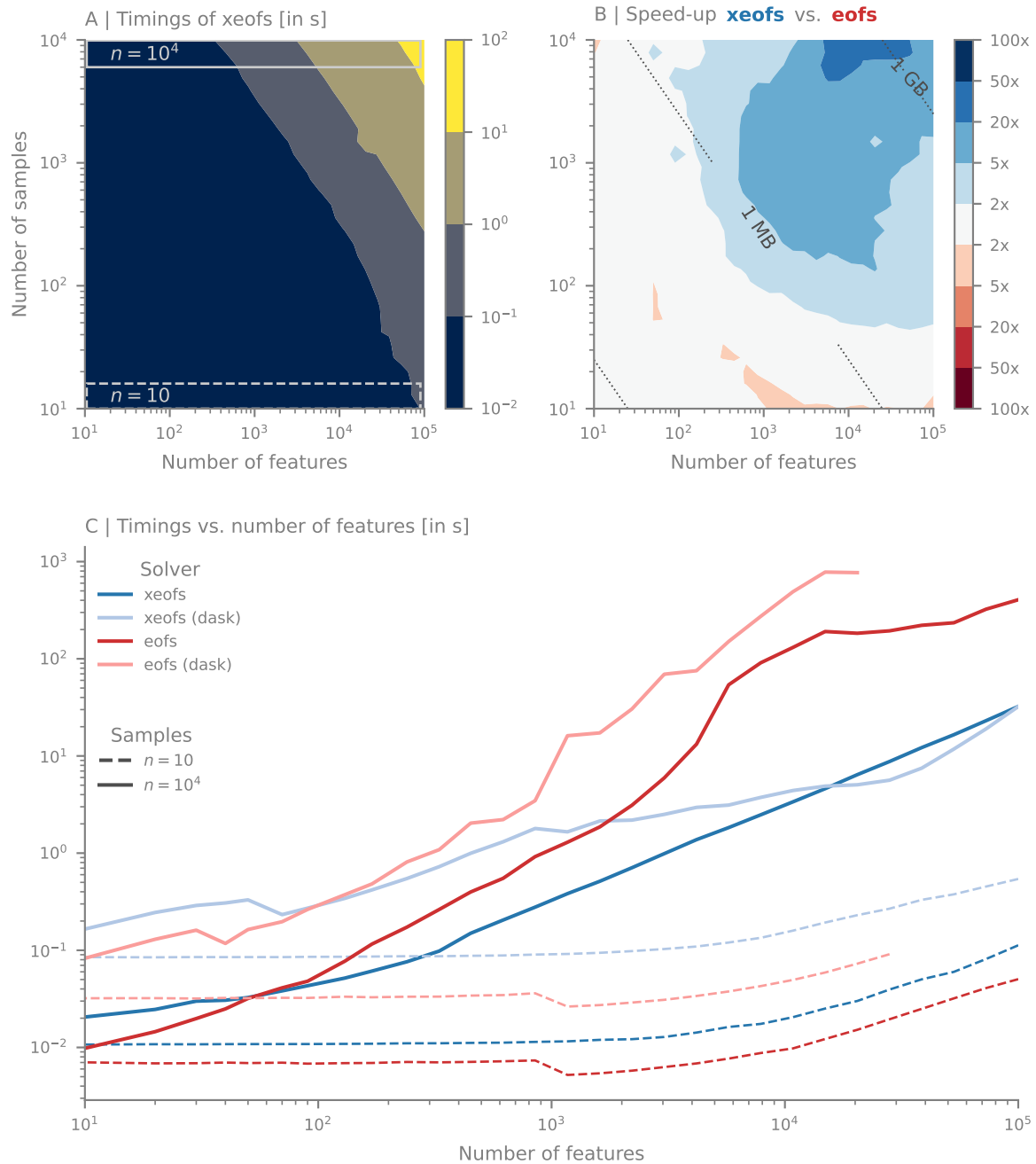


Figure 2.2 — Computational performance of PCA using the XEOFS library (version 3.0.0). Panel A displays computation times (in seconds) for varying dataset sizes, with different numbers of observations (samples) and variables (features). Panel B illustrates the speedup achieved by XEOFS compared to the EDFS library (version 1.4.1). Panel C presents the performance of delayed (lazy) Dask computation versus eager computation in both small data ($n = 10$ samples) and large data ($n = 10,000$ samples) scenarios. The results demonstrate a significant efficiency gain for medium to large datasets (greater than 1 MB) where XEOFS outperforms EDFS by approximately an order of magnitude. Additionally, XEOFS's Dask implementation for chunked out-of-memory data offers further improvements for larger datasets. Interestingly, this efficiency gain is observed primarily for datasets with a balanced number of features and samples, while performance does not significantly improve for extremely "tall-and-skinny" or "short-and-wide" matrices. Notably, the Dask implementation in EDFS performs worst on larger datasets. For small datasets (less than 1 MB), the Python overhead is more pronounced, with EDFS being slightly more efficient, although the difference is generally less than a factor of two. The benchmark was conducted on a system with Dual Intel Xeon Gold 6248R 24C 3.0 GHz processors and 192 GB of RAM, running Rocky Linux 8.

30 Dimensionality Reduction Techniques in Climate Science: Bridging Theory and Practice

1 Future updates to XEOFS could introduce additional state-of-the-art methods that have recently
2 gained attention. Nonlinear dimensionality reduction techniques, such as those proposed by
3 Bueso *et al.* [13] as well as spectral methods as proposed by Guilloteau *et al.* [44], would enable
4 researchers to capture more complex relationships within climate datasets. Incorporating
5 approaches like nonlinear PCA, kernel-based methods [106], and advanced sparse algorithms
6 to improve interpretability [43, 107] would provide users with a broader toolkit to explore the
7 intricate dynamics often present in climate data.

8 Another area for improvement is the expansion of bootstrapping techniques for evaluating
9 the statistical significance of extracted patterns, particularly in MCA and rotated PCA. Boot-
10 strapping helps researchers assess the robustness of their findings, distinguishing meaningful
11 patterns from noise. By integrating more advanced bootstrapping algorithms, such as those
12 discussed by Timmerman *et al.* [113], XEOFS could simplify the workflow for significance
13 testing, making these analyses more accessible without requiring additional coding expertise.

14 The next chapter will focus on applying dimensionality reduction techniques to real-world
15 climate datasets, highlighting the practical use of XEOFS. These applications will demonstrate
16 how the package can help extract meaningful insights from complex datasets. by exploring
17 interactions between the ocean and atmosphere.

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Proofs

The Frobenius norm $\|\cdot\|_F$ of a complex $N \times P$ matrix $\mathbf{A} \in \mathbb{C}^{N \times P}$ is defined as:

$$\|\mathbf{A}\|_F \equiv \left(\sum_i^N \sum_j^P |\alpha_{ij}|^2 \right)^{1/2} = \left(\sum_i^{\min(N,P)} \sigma_i^2(\mathbf{A}) \right)^{1/2}. \quad (\text{A.1})$$

This expression reveals that the Frobenius norm is equivalent to the ℓ_2 -norm of the matrix's singular values σ_i . We can demonstrate this equivalence using the singular value decomposition (SVD) of $\mathbf{A}$ and leveraging the property that the trace of a matrix remains invariant under cyclic permutations.

Starting with the definition of the Frobenius norm, we have:

$$\|\mathbf{A}\|_F = \left(\text{tr}(\mathbf{A}^H \mathbf{A}) \right)^{1/2}$$

Applying the SVD, we can express $\mathbf{A} = \mathbf{V} \mathbf{\Sigma} \mathbf{V}^H$, leading to:

$$\|\mathbf{A}\|_F = \left(\text{tr}(\mathbf{V} \mathbf{\Sigma} \mathbf{U}^H \mathbf{U} \mathbf{\Sigma}^H \mathbf{V}^H) \right)^{1/2}$$

Since $\mathbf{U}^H \mathbf{U} = \mathbf{I}$, this simplifies to:

$$\|\mathbf{A}\|_F = \left(\text{tr}(\mathbf{V} \mathbf{\Sigma}^2 \mathbf{V}^H) \right)^{1/2}$$

Finally using the invariance property of the trace, we obtain:

1

$$\|\mathbf{A}\|_{\text{F}} = \left(\sum_i^{\min(N,P)} \sigma_i^2(\mathbf{A}) \right)^{1/2}$$

- 2 This concludes the proof that the Frobenius norm of a matrix is indeed the ℓ_2 -norm of its
3 singular values.

